Aaryuman Sawai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.75
St Xavier's Residential School Mandi	Class XII (CBSE)	2020	91.00%
DAV Centenary Public School Jawahar nagar mandi hp	Class X (CBSE)	2018	93.00%

SCHOLASTIC & COMPETITIVE PROGRAMMING ACHIEVEMENTS

- Solved 500+ problems on LeetCode in C++, including 171 Easy, 270 Medium and 61 Hard problems; earned the 365 Days Badge.
- Built strong problem-solving proficiency across Dynamic Programming, Graphs, BFS/DFS, Backtracking, Divide and Conquer, Greedy Algorithms, Hash Tables, Arrays, Strings, Sorting, Binary Search and Recursion.
- Maintained 250+ active days and 1,200+ submissions on LeetCode over the displayed one-year period.
- LeetCode contest rating: 1,431; participated in 22 contests.
- Qualified NTSE Stage 1 from Himachal Pradesh.

TECHNICAL SKILLS

Languages: C++, C, Java, Python, JavaScript, SML

DSA: Data Structures, Algorithms, Competitive Programming, Dynamic Programming, Graphs, BFS, DFS, Backtracking, Greedy, Divide and Conquer, Hashing, Binary Search, Recursion

Web Development: HTML, CSS, JavaScript, React, Svelte, Node.js, Express.js, EJS

Data Science / ML: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Jupyter Notebook, EDA, Data Cleaning, Feature Engineering, Statistical Analysis, Data Visualization, K-Means, PCA

Databases: PostgreSQL, SQL, MongoDB, Mongoose

Tools: Git, GitHub, GitLab, Bitbucket, VS Code, Eclipse, Unix/Linux

Core CS / Other: Object-Oriented Programming, Compiler Design, MQTT, IoT Security, Encryption, Cryptography, Flex, Yacc

EXPERIENCE

Full Stack Developer Intern -- Atoll Solutions Private Limited | May 2023 - Jul 2023

- Developed full-stack web applications using React, Svelte, Node.js and MongoDB.
- Implemented backend functionality for a contact-list application using Node.js and MongoDB.
- Built multiple React single-page applications to strengthen frontend development and component-based UI skills.
- Evaluated React and Svelte for development suitability and prepared a comparative report.

PROJECTS

Netflix Movies & TV Shows -- Data Analysis -- Python | Pandas | NumPy | Matplotlib | Seaborn | Jupyter Notebook

- Performed exploratory data analysis on the Netflix Movies and TV Shows dataset using Pandas and NumPy to identify trends in content type, release years, genres, countries and ratings.
- Cleaned and transformed raw data by handling missing values, removing duplicates, converting date fields and extracting year-based features for analysis.
- Created visualizations using Matplotlib and Seaborn to analyze content growth, genre distribution, ratings, country-wise distribution and release trends.
- Derived insights using grouping, filtering, aggregation and statistical analysis.

Customer Segmentation & Exploratory Data Analysis -- Python | Pandas | NumPy | Matplotlib | Seaborn | scikit-learn | K-Means | PCA

- Performed end-to-end exploratory data analysis and preprocessing on customer data using Pandas and NumPy, including data cleaning, missing-value handling, feature preparation and statistical exploration.
- Applied K-Means clustering to identify customer segments and used PCA for dimensionality reduction and visualization of cluster structure.
- Created visual analyses using Matplotlib and Seaborn to communicate distributions, relationships and customer-segment patterns.

Jarvis AI -- Python | Google Gemini API | Speech Recognition | Windows SAPI

- Developed a Python-based voice assistant capable of accepting voice commands and generating AI-powered responses using the Google Gemini API.
- Integrated speech_recognition for voice input and Windows SAPI (SAPI.SpVoice) for text-to-speech output.
- Built a modular Gemini API integration to connect the assistant with Google's generative AI services.

Bank Management System -- PostgreSQL | Frontend | Backend

- Developed a bank-management application with PostgreSQL database integration and frontend/backend components.

Compiler Design Project -- Flex | Yacc

- Designed and implemented a functional compiler for a compiler-design laboratory project using Flex and Yacc.

Contact List -- Node.js | Express.js | EJS | MongoDB | Mongoose

- Developed a full-stack contact-management application for adding and managing contact information.
- Implemented server-side functionality and MongoDB persistence using Express.js and Mongoose.

Movie Flix -- Web Development

- Built a Netflix-style movie/streaming website.

Authentication in IoT Devices Using MQTT -- MQTT | Encryption | IoT Security

- Implemented encryption methods for secure communication between IoT devices using the MQTT protocol.
- Developed as an undergraduate project under Dr. Vivek Chaturvedi.

POSITIONS OF RESPONSIBILITY

- Placement Coordinator, IIT Hyderabad Placement Cell -- 2026-27 session.
- OCS Volunteer / Placement Proctor, IIT Hyderabad.
- Web Developer, Sunshine Club, IIT Hyderabad -- Student Club.
- CDC Coordinator, IIT Palakkad -- coordinated career-development activities for 2nd-year CSE students.
- Organized a crypto-trading competition and coordinated participation and event execution.
- NSS volunteer; organized a blood-donation camp and motivated 100+ participants.

TRAINING

- Harvard CS50 -- May 2024.
- Introduction to Java -- Coding Ninjas, Oct 2022.
- Data Structures & Algorithms -- Coding Ninjas, Sep 2021.

HOBBIES & INTERESTS

Chess | Learning new languages